An Improved MODIS NIR PWV Retrieval Algorithm Based on an Artificial Neural Network Considering the Land-Cover Types

Xiongwei Ma, Yibin Yao, Bao Zhang, Yuxin Qin, Qi Zhang, and Hang Zhu

***Abstract*— Estimating precipitable water vapor (PWV) with high accuracy and spatial resolution is important in many disciplines. Water vapor absorption and non-absorption channels can be observed in the near-infrared (NIR) ray of the Moderate Resolution Imaging Spectroradiometer (MODIS), which can be used to retrieve PWV. However, traditional algorithms overestimate the NIR PWV in North America. This study proposes a novel NIR retrieval algorithm based on machine learning that considers land-cover types to estimate high-accuracy PWV. To do this, nonlinear models between MODIS NIR transmittance, based on the two-and three-channel ratio, and global navigation satellite system (GNSS) PWV, recorded by the SuomiNet GNSS network, are established using a backpropagation neural network (BPNN). Verification shows that the root mean square error (RMSE)/standard deviation (STD)/bias of the two-channel ratio PWV is 1.29/1.29/0.02 mm, respectively, and the improvements of RMSE and STD are 66.32% and 37.98%, respectively. The RMSE/STD/bias values of the three-channel ratio PWV are 1.29/1.29/0.02 mm, respectively, and the improvements in RMSE and STD are 68.67% and 42.31%, respectively. In addition, the surface verification of the proposed method in six land-cover types shows that both the two-and three-channel ratio methods can yield satisfactory PWV estimates. Compared with the MODIS PWV products, the proposed method yields remarkable progress.**

***Index Terms*— Global navigation satellite system (GNSS), precipitable water vapor (PWV), remote sensing, retrieval.**

I. Introduction

ATMOSPHERIC water vapor, also called precipitable water vapor (PWV), is the most active and abundant component of the climate system and a critical carrier in the hydrological cycle and energy transformation between the surface and the atmosphere [1], [2]. Water vapor evaporates from the surface or ocean, condenses to form clouds, and is transported elsewhere, eventually falling to the Earth's surface during precipitation [3]. Water vapor also affects the atmospheric environment because it affects the size, composition, and optical properties [4], [5]. In addition, PWV can also calibrate the wet delay of geodetic observations, improving the accuracy of satellite geodesy and altimetry [6], [7]. Therefore, accurate PWV measurement is key to improving the understanding of the water cycle, monitoring the atmospheric environment, and calibrating various geodetic observation errors. However, because the spatiotemporal variation in PWV is complex, it is challenging to measure it accurately.

Although there are several methods to observe PWV, including water vapor radiometer, radiosonde, and solar photometer [8], [9], the low spatiotemporal resolution leads it impossible to understand the variation and distribution of PWV accurately. In recent decades, global navigation satellite system (GNSS) meteorology has made remarkable progress, making GNSS a powerful means of observing PWV with high precision and temporal resolution [10]–[12]. Moreover, GNSS PWV has been used as a ground reference in many studies [13], [14]. Numerical weather prediction models, such as European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis 5 (ERA5), can also simulate PWV. However, ERA5 provides PWV hourly with a spatial resolution of 0.25°× 0.25° (about 31 km × 31 km) [15]. For small areas, the coarse spatial resolution of PWV—0.25° × 0.25°—still needs to be improved to obtain detailed spatial variations [16], [17]. With the improvement of high-resolution sensors and computer storage and processing capacity, space-based remote sensing technology can continuously observe regional and global PWV distribution, providing an effective way to obtain high-resolution PWV [16], [17].

PWV can be estimated from two types of remote sensing data: optical remote sensing observations [infrared (IR) and near-infrared (NIR)] [18] and microwave observations (MW) [19], [20]. IR and MW can measure PWV under all-weather conditions with a coarse spatial resolution [21], [22]. NIR is more sensitive to the boundary layer where PWV is located, and PWV products can be obtained with a 1-km resolution [23]. Among various satellite-based PWV products, the Moderate Resolution Imaging Spectroradiometer (MODIS) onboard the Aqua (launched on December 18, 1999) and Terra (launched on May 4, 2002) can provide global observations for comprehensive scientific research for land, oceans, and

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atmosphere; it is also the first space instrument to observe PWV through IR and NIR channels with a spectral range of 0.4–14.4 μm [23], [24]. Five NIR channels, including three water vapor absorbing channels centered near 0.905, 0.936, and 0.940 μm and two atmospheric window channels at 0.865 and 1.24 μm, can be used to observe PWV. The 1 km × 1 km high spatial resolution PWV product (MOD05_L2) can be derived based on the NIR retrieval algorithm [18]. Regions with NIR reflective surfaces, including clear-sky land, clouds, and ocean with sunlight, can provide NIR PWV products [25] with 20% uncertainty, significantly overestimating the PWV value [26], [27].

In recent years, two ways to refine the accuracy of MODIS NIR PWV have predominated: one is to calibrate NIR PWV products, and the other is to refine the NIR PWV retrieval algorithm. Regarding the calibration, Liu *et al.* [28] evaluated the MODIS NIR PWV products using the 83 radiosonde PWV in China and corrected them with a linear fitting model. Bai *et al.* [29] developed a linear calibration model for the MODIS PWV considering regional climate differences. The calibrated PWV reduced the rms by approximately 14.7%, 17.7%, 2.9%, 2.5%, and 11.3% in five different climatic regions, respectively. Zhang and Yao [30] proposed a generalized regression neural network method to calibrate California MODIS NIR PWV; the improved rms was 2.2 mm. Although the optimization method can improve the accuracy of MODIS PWV, it is an indirect method based on PWV products. Another method is to improve the NIR retrieval algorithm using the original observation data to improve the accuracy of PWV. For example, Merrikhpour and Rahimzadegan [31] improved the NIR PWV accuracy by optimizing the ratio of water vapor absorbing channels and atmospheric window channel coefficients, modeling the bias between NIR and the radiosonde PWV. Mao *et al.* [32] constructed an artificial neural network (ANN) model under different weather conditions to retrieve the NIR PWV, but they did not consider the uncertainty caused by uneven surface-type characteristics. He and Liu [13] established the relationship between GNSS PWV and transmittance using a regression fitting method considering the surface type; however, this does not adequately consider the nonlinear relationship between PWV and transmittance and has poor accuracy on some surfaces.

A nonlinear NIR PWV retrieval model considering surface cover types is proposed in this study to reduce the uncertainty of the NIR PWV retrieval algorithm caused by the nonlinear relationship between atmospheric water vapor and transmittance and land-cover types. An ANN was successfully applied in developing retrieval algorithms computationally for satellite data and quantitative remote sensing [14], [33], [34]; they are specially designed to capitalize on the inherent statistical relationship between the retrieved parameters and are very suitable for solving nonlinear problems [33]. In this study, ANN technology was used to establish the relationship between PWV observed by the SuomiNet GNSS rather than simulated and transmittance. For verification, approximately 20% of PWV samples did not participate in model construction and were used to verify the retrieved NIR PWV. Section II introduces the research area and data used in the retrieval algorithm. Section III introduces the basic theory and estimation method of the proposed algorithm. The fourth part is the verification and result part of the algorithm, and Section V summarizes this research.

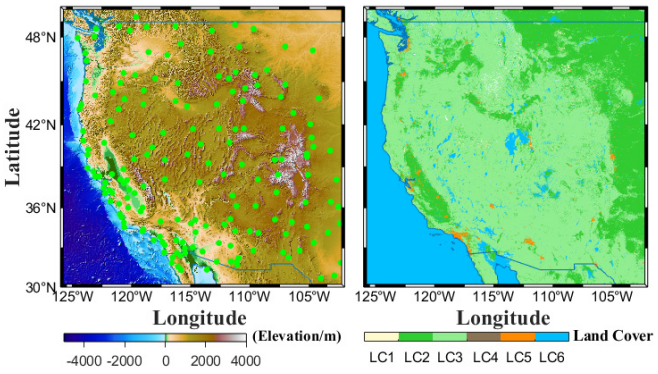


Fig. 1. Study area. (Left) Map shows the distribution of GNSS sites and the topography of North America. (Right) Map shows the land cover of the research area.

II. Study Area and Data

A. Study Area

North American land was selected as the research area bounded by 30°N–50°N and 126°W–102°W. Fig. 1 (left) shows the topography and the distribution of the GNSS sites in this study. The topography varies from a few meters below zero near the east coast of the Pacific Ocean to approximately 4000 m near the center of the study area. A total of 173 GNSS stations were distributed at different altitudes. The varied topography and the migration of land–sea water vapor led to severe spatial variations in PWV in the study area. Fig. 1 (right) shows the land cover of the study area. LC1–LC6 represents six land-cover types (LC1, farmland; LC2, woodland; LC3, grassland; LC4, bare land; LC5, urban area; and LC6, water body). The surface cover was mainly LC2 and LC3, accounting for 78.37% of the study area.

B. Data

1) Ground-Based GNSS PWV and ERA5 PWV: The ground GNSS PWV was derived from the SuomiNet Network and is accessible at https://data.cosmic.ucar.edu/suominet/postProcess/pwvConus/. SuomiNet is a proposed network of GNSS receivers that provides postprocess and real-time atmospheric PWV measurements, and geodetic and meteorological information [35]. SuomiNet data are provided in various formats to access raw and derivative data products. Most GNSS sites are co-located with meteorological stations, whereas others use meteorological data near automatic surface observation system sites. Meteorological data were used to calculate the zenith hydrostatic delay (ZHD) on the GNSS sites, which can be expressed as [36]

$$\mathrm{ZHD} = \frac{0.002277 \times P_s}{1 - 0.00266 \times \cos(2\varphi) - 0.00028 \times H} \tag{1}$$

where P_s refers to the surface pressure (hPa). φ and H are the latitude (rad) and altitude (km), respectively.

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TABLE I
SPECTRAL POSITION AND INFORMATION OF MODIS NIR CHANNEL FOR WATER VAPOR DETECTION

Band Number	Band Width (nm)	Center wavelength	Spatial resolution (m)	Description
2	841-876	865	250	Window Channel
5	1230-1250	1240	500	Window Channel
17	890-920	905	1000	Absorption Channel
18	931-941	936	1000	Absorption Channel
19	915-945	940	1000	Absorption Channel

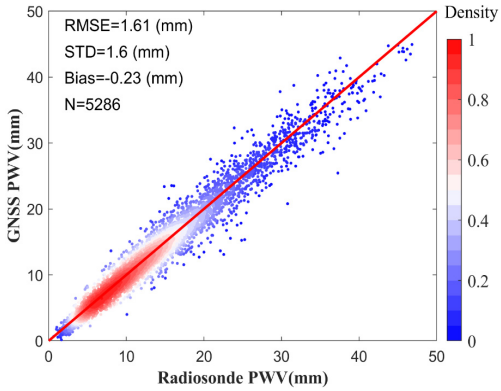


Fig. 2. GNSS PWV versus radiosonde PWV in study area.

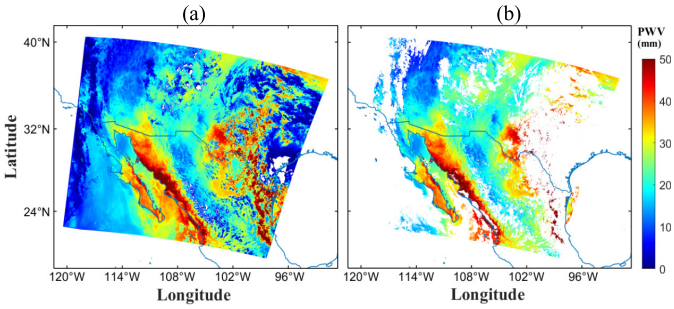


Fig. 3. MODIS NIR PWV distribution on the 153th day in 2020, 17:55(UTC). (a) and (b) PWV distribution under cloudy and clear conditions, respectively.

Zenith wet delay1 (ZWD) can be obtained by removing ZHD from the zenith total delay (ZTD), and then PWV can be obtained by the conversion parameters Π

$$\text{ZWD} = \text{ZTD-ZHD} \tag{2}$$

$$\text{PWV} = \Pi \times \text{ZWD} \tag{3}$$

where Π can be obtained using the following equation:

$$\Pi = \frac{10^5}{(k_3/T_m + k_2')R_v} \tag{4}$$

where T_m refers to the atmospheric weighted average temperature (K), which can be obtained based on the empirical model using the observed surface temperature [37]. $k_2' = 6.25$ (K/hPa), $k_3 = 3.776 \times 10^5$ (K^2/hPa), and $R_v = 461.495$ (J/K/kg) are constants. In this study, the postprocess SuomiNet PWV files are updated daily and provide estimated PWV temporal resolution of 30 min. Compared with the observed and co-located radiosonde data, the rms/standard deviation (STD)/bias of SuomiNet PWV products is 1.61/1.60/−0.23 mm, respectively, which shows good reliability (see Fig. 2).

ERA5 is the fifth-generation global climate reanalysis of ECMWF, which provides the PWV with an accuracy of about 2 mm [38]. The hourly ERA5 PWV data with horizontal resolution of 31 km in 2020 are used in this study and are freely accessible at (https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5).

2) Optical Remote Sensing Data: Four optical remote sensing data were used in this study: MOD021KM, MOD03, MOD35, and MOD05_L2. The original data can be accessed at https://ladsweb.modaps.eosdis.nasa.gov/, stored as hierarchical data format (HDF) format in sinusoidal projection. MOD021KM contains calibrated and geolocated at-aperture radiances for 36 discrete bands. In this study, two atmospheric window channels (Band 02 and Band 05) and three atmospheric absorption channels (Band 17, Band 18, and Band 19) were used to calculate water vapor transmission and propose an NIR PWV retrieval algorithm. The details of these five bands are listed in Table I. It should be noted that the original band data are stored with the digital number (DN); therefore, the DN value should be converted into zenith reflectivity using the radiometric calibration equation

$$\text{Reflectance} = \text{scale} \times (\text{DN} - \text{offset}) \tag{5}$$

where scale and offset are the scale and offset factors, respectively. MOD03 is the geolocation data used for the geometric correction of the five bands and MOD05_L2 NIR PWV products. MOD35 is a cloud-mask product. As NIR ray cannot penetrate thick cloud, the cloud area should be identified and removed. The level 2 NIR PWV product with a fine spatial resolution of 1 km × 1 km was derived from MOD05_L2. Fig. 3 shows a PWV map that was calibrated geometrically. Fig. 3(a) shows the MOD05_L2 NIR PWV map at 17:55 (UTC) on the 153rd day in 2020 with clouds, while Fig. 3(b) shows a clear sky. Clouds have an impact on NIR PWV detection.

3) Digital Elevation Model (DEM) and Land Cover: The DEM product was derived from the Shuttle Radar Topography Mission (SRTM) with a resolution of 30 m; The digital elevation model (DEM) product, derived from the SRTM with a resolution of 30 m, is resampled to 1000 m by the China Resource and Environmental Science and Data Center (RESDC) and

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is available for free (http://www.resdc.cn/). It is well-known that the GNSS heights are ellipsoid, while the SRTM DEM heights are orthometric. Hence, to unify the height systems, we use the undulations from the EGM2008 model to get rid of the differences between the ellipsoidal height and the orthometric height. The land-cover data were provided by the European Space Agency Climate Change Initiative (ESACCI) (http://maps.elie.ucl.ac.be/CCI/viewer/index.php) with a spatial resolution of 300 m. The original product was divided into 22 categories, sampled to 1000 m and classified as farmland, woodland, grassland, bare land, urban area, and water bodies (LC1–LC6) in this study.

III. Methodology

A. Physics Background

1) MODIS PWV Retrieval Principle: The reflectivity of the Earth's atmosphere is affected by the physical properties of the air molecules, where asymmetric molecules, such as water molecules, affect the transmission of solar radiation [39], [40]. Therefore, atmospheric water vapor content can be retrieved from the ratio of the solar radiation reflected by the surface in the absorption channel and the adjacent non-absorption channel. The solar radiation between 0.86 and 1.24 μm in the "Sun–surface–sensor" is affected by water vapor absorption [41]. The radiation luminance of the entrance pupil of the satellite sensor can be expressed as follows [18]:

$$L_{\text{Sensor}}(\lambda) = L_{\text{Sun}}(\lambda)T(\lambda)\rho(\lambda) + L_{\text{Path}}(\lambda) \tag{6}$$

where λ refers to the wavelength, $L_{\text{Sensor}}(\lambda)$ refers to the radiance of the entrance pupil at the sensor, $L_{\text{Sun}}(\lambda)$ refers to the solar radiance, $T(\lambda)$ refers to the total transmittance on the optical slant path, $\rho(\lambda)$ refers to the surface reflectance, and $L_{\text{Path}}(\lambda)$ refers to the path-scattered radiance. The path-scattered radiance can be ignored under a clear sky and high visibility. Therefore, (6) can be expressed as

$$\rho^*(\lambda) = \frac{L_{\text{Sensor}}(\lambda)}{L_{\text{Sun}}(\lambda)} = T(\lambda)\rho(\lambda) \tag{7}$$

where $\rho^*(\lambda)$ is the reflectivity observed from the sensor. The relationship between transmittance and water vapor content can be simulated by MODTRAN and LOWTRAN [42]

$$T(\lambda) = \exp\left(\alpha - \beta\sqrt{w}\right) \tag{8}$$

where w is the water vapor content over its optical slant path, and α and β are constants.

2) Transmittance Calculation Using Differential Absorption Techniques: As the effect of different underlying surfaces on the reflectivity of the same wavelength is different, transmittance cannot be obtained from the radiation of a single channel, which is generally calculated by the two-or three-channel ratio. Assuming that the surface reflectivity does not change with the wavelength, the transmittance can be obtained from the ratio of one absorption channel to one window channel called the two-channel ratio method

$$T_i(\lambda) = \frac{\rho_i}{\rho_2} \tag{9}$$

where $T_i(\lambda)$ refers to the transmittance for band i (i = 17, 18, 19), ρ_i refers to the reflectance of band i, and ρ_2 refers to the reflectance of the window channel (Band 2). If the surface reflectivity varies linearly with the wavelength, the transmittance can be obtained from the ratio of one absorption channel and two window channels called the three-channel ratio method

$$T_i(\lambda) = \frac{\rho_i}{C_1 \cdot \rho_2 + C_2 \cdot \rho_5} \tag{10}$$

where C_1 and C_2 refer to constants with values of 0.8 and 0.2, respectively, [43]; ρ_2 and ρ_5 are the reflectance values of Band 2 and Band 5, respectively.

B. Model Development

1) ANN: ANNs are quasi-human brain biological neural networks, including input, hidden, and output layers [44]. The backpropagation neural networks (BPNNs), one of the most successful network learning algorithms, are widely applied [45]. BPNN is usually designed with three layers: input, hidden, and output. Each layer is composed of several nonlinear processing units, and the adjacent layers are connected by a synaptic weight matrix [46], [47]. BPNN is characterized by the forward transmission of the signal and the backpropagation of the error. In forward propagation, the signal is transmitted unidirectionally to the input layer, processed in the input and hidden layers, and finally transmitted to the output layer. If the error between the output and the target value is beyond the range of the allowable error, backpropagation will be activated to adjust the weight so as to reduce the error [48], [49]. In principle, if there are sufficiently large samples, the nonlinear relationship between the input and the output can be accurately fitted by BPNN. Although there is a significant relationship between atmospheric water vapor and transmittance, which is complex and nonlinear, it is difficult to simulate using accurate physical equations. Therefore, BPNN was selected to establish a novel MODIS NIR PWV retrieval algorithm.

2) Methodology for MODIS NIR PWV Retrieval: A nonlinear model was used to establish a high-precision MODIS NIR PWV retrieval model to establish a nonlinear functional model between the measured PWV and atmospheric water vapor transmittance. As GNSS observes the total water vapor content in the zenith direction, it should be projected onto the optical slant path (Sun–surface–sensor)

$$\text{PWV}^* = \text{PWV}\left(\frac{1}{\cos\theta} + \frac{1}{\cos\gamma}\right) \tag{11}$$

where PWV refers to the GNSS PWV in the zenith direction, and PWV* refers to PWV on the optical slant path projected by GNSS PWV. θ and γ refer to the sensor and solar zenith angle, respectively, which can be obtained from the MOD03 datasets. As machine learning is excellent in establishing a model that is complex and difficult to express, a BPNN was used to build the retrieval model. In addition, the sensitivities of the three water vapor absorption channels were different. Fig. 4 shows the relationship between PWV on the optical slant path obtained based on GNSS and transmittance using the two-and three-channel ratio methods for different land-cover

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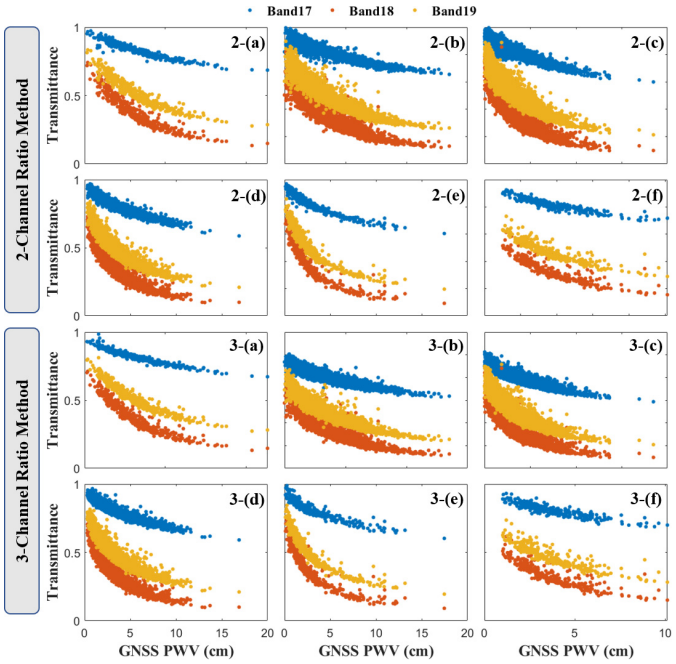


Fig. 4. Relationship between two-and three-channel-ratio-based transmittance and GNSS PWV on the optical slant path (Sun–surface–sensor) in different land-cover types. "Two" and "three" refer to the transmittances calculated by the two-channel ratio method and three-channel ratio method, respectively. (a)–(f) Six land-cover types.

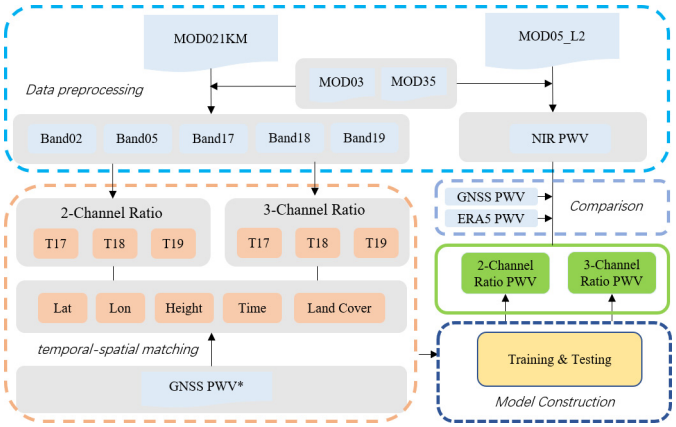


Fig. 5. Entire workflow of the proposed methods.

types. It was observed that band 18 was the most sensitive to PWV, and the transmittance decreased drastically with an increase in PWV. Band 17 is the least sensitive to PWV, and its transmittance variation was the smallest. Band 19 has moderate transmittance. This result is consistent with [13]. Therefore, the estimated PWV from the three bands is expected to have a better performance than that from a single absorption channel. In addition, the surface of the Earth is highly heterogeneous. Different land-cover types have different effects on the reflectivity of the same wavelength. Fig. 4 indicates that the relationship between the atmospheric absorption channels and GNSS PWV on different surfaces is different. Therefore, land-cover types should be considered when constructing retrieval models.

Fig. 5 shows the workflow of the proposed methods. First, preprocessing of the optical remote sensing data is necessary. The preprocessing of MOD05_L2 includes NIR PWV collection, geometric calibration, and cloud processing. The preprocessing of MOD021KM includes band collection under a clear sky, radiometric calibration, and geometric calibration. The atmospheric water vapor transmittance was then calculated. There are two methods for obtaining transmittance, including the two-and three-channel ratio methods. Therefore, the PWV retrieval models are based on the two-and three-channel ratio methods using the BPNN algorithm to compare the performance of retrieval models proposed by different methods.

As the GNSS and optical remote sensing data have different spatiotemporal resolutions, the mismatch problem needs to be addressed. The temporal resolution of GNSS PWV is 30 min, and optical remote sensing data are 5-min products. Therefore, the matched time differences were controlled within 30 min for ground-based GNSS sites and MODIS data. Specifically, the GNSS time is selected as a reference to search the MODIS data in the pre-and-post 15 min. If the MODIS data meet the criteria, the qualified MODIS data are averaged to form a GNSS–MODIS pair. As for the spatial mismatch, dense MODIS data were interpolated to sparse GNSS sites using a linear interpolation method to obtain the average values of four pixels around the sites. As for land-cover type and DEM, only the spatial mismatch problem needs to be addressed. The dense land-cover type and DEM data were interpolated to sparse GNSS sites using a nearest-neighbor interpolation method. Finally, 28 272 GNSS–MODIS/DEM/LC pairs in 2020 were obtained.

The 28 272 pairs of samples were used as the learning samples to construct the BPNN models described in Section III-A1. As two different transmittances can be obtained based on the two-and three-channel ratio methods, two BPNN models were constructed in this study. When the input transmittance is calculated based on the two-channel ratio method, we call the corresponding BPNN the two-channel ratio model; when the input transmittance is calculated based on the three-channel ratio method, we call it the three-channel ratio model. The structures of the two models were similar. Fig. 6 shows that the input layer is composed of eight neurons. The observation vector comprises latitude, longitude, height, time, land cover, and three atmospheric water vapor transmittances (based on the two-and three-channel ratio methods, respectively), where $X^i = [\text{Latitude Longitude Height Time LC } T17\ T18\ T19]^T, i < 1 \leq 28\,272$. The output layer is one neuron, which is the total water vapor content of the optical slant path based on GNSS PWV projection, $Y^i = [\text{PWV}^*_{\text{GNSS}}]$. A single hidden layer BPNN model is selected in this study, and the method for determining the number of nodes in the hidden layer can be found in Section III-B3. The aim of the BPNN model is to establish a functional mapping relationship $f_{2-\text{Channel}}$ and $f_{3-\text{Channel}}$ between the input and output data. When $f_{2-\text{Channel}}$ and $f_{3-\text{Channel}}$ are obtained, the two-and three-channel ratio PWV over its optical slant path can be obtained by recording $X = [\text{Latitude Longitude Height Time LC } T17\ T18\ T19]^T$. The two-and three-channel ratio PWV in the vertical direction (11) can be obtained using the sensor zenith angle and solar zenith angle.

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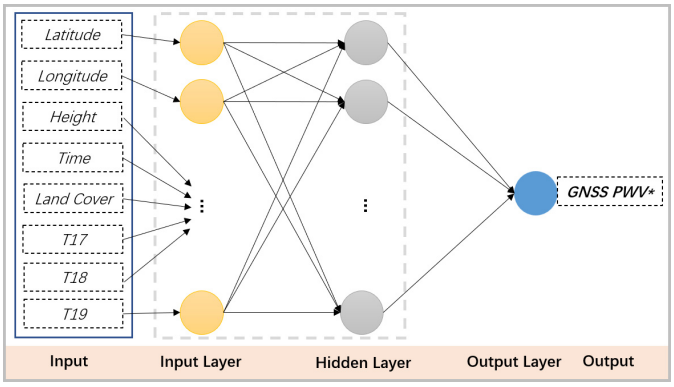


Fig. 6. Architecture of the BPNN model designed for MODIS NIR PWV retrieval.

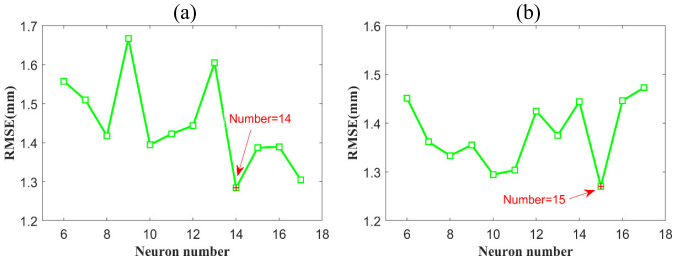


Fig. 7. Comparisons of RMSE using different neuron numbers in the hidden layer for the BPNN model of (a) two-channel ratio PWV and (b) three-channel ratio PWV. The red cross marks the minimum rms.

3) Determining the Neuron Number in the Hidden Layer: How the BPNN performs significantly depends on the proper number of neurons in the hidden layer. On one hand, the limited hidden neurons might cause the underfitting problem. On the other hand, excessive neurons in the hidden layer will not only cause the overfitting phenomenon [50] but also affect the efficiency of the training model. In this contribution, we chose *a posteriori* method to fix the optimal hidden neurons. Previous studies [51]–[53] have found that the neurons in the hidden layer generally range from $2(N)^{1/2}+\upsilon$ to $2N+1$, where N and υ are the number of nodes in the input and output layers, respectively. In this article, N and υ, respectively, equal to 8 and 1 both in the two-and three-channel ratio models, since there is the same number of input and output neurons in the two training models. Therefore, we repeatedly trained the two BPNN models as the hidden neurons changed from 6 to 17. In each experiment, we randomly chose 80% of the data pairs to train the models, while the rest of the data (approximately 20%) were applied to verify the effectiveness of the models. The optimal hidden neurons in the two models were evaluated by the root mean square error (RMSE) of the verification dataset. Fig. 7 illustrates the RMSE with different hidden neurons in the two BPNN models of the two-and three-channel ratio PWV. It was evident that the optimal hidden neurons in the two-and three-channel ratio PWV retrieval models were 14 and 15, respectively, corresponding to the minimum RMSE of the two models.

C. Result Validation

The bias, STD, RMSE, and correlation coefficient (R) between the two-and three-channel ratio PWV and GNSS PWV at all the sites were calculated and used as indicators to test the performance of the two-and three-channel ratio PWV retrieval models based on BPNN. Moreover, ERA5 PWV is also used to verify the results of the proposed methods. The equations for the four indicators are as follows, (12)–(15), as shown at the bottom of the next page, where N refers to the number of samples, $\mathrm{PWV}_i^{\mathrm{pre}}$ is the PWV values' output from the BPNN models, and $\mathrm{PWV}_i^{\mathrm{obs}}$ are the reference PWV values. $\overline{\mathrm{PWV}_i^{\mathrm{pre}}}$ and $\overline{\mathrm{PWV}_i^{\mathrm{obs}}}$ are the mean values of $\mathrm{PWV}_i^{\mathrm{pre}}$ and $\mathrm{PWV}_i^{\mathrm{obs}}$, respectively.

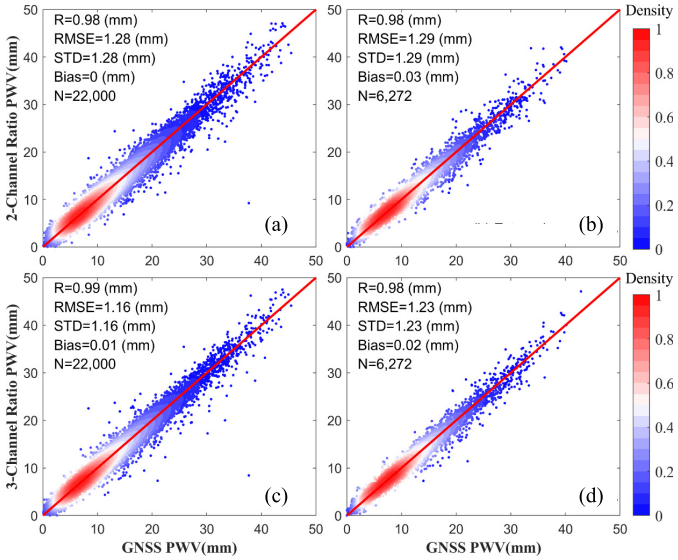


Fig. 8. Two-and three-channel PWV versus GNSS PWV at (a) and (c) training sites and (b) and (d) test sample sites, respectively.

IV. Results

A. Overall Performance

To verify the performance of the retrieval model, the scatter plots show the two-and three-channel ratio PWV versus GNSS PWV in North America at the training and test sample sites (see Fig. 8). It is observed that the two-and three-channel ratio models are robust, and the obtained PWV has suitable accuracy. The internal and external R/RMSE/STD/bias of the two-channel ratio model are 0.98/1.28 mm/1.28 mm/0 mm and 0.98/1.29 mm/1.29 mm/0.03 mm, respectively. The internal and external accuracy of the three-channel ratio model was better than that of the two-channel ratio model, with R/RMSE/STD/bias values of 0.99/1.16 mm/1.16 mm/0.01 mm and 0.98/1.23 mm/1.23 mm/0.02 mm, respectively. The bias of PWV generated by the two-and three-channel ratio model is equal to 0, indicating that the model based on BPNN is unbiased with reference to GNSS PWV. There are no systematic biases between the obtained PWV and GNSS PWV. In addition, the accuracy of the two-and three-channel ratio models and NIR PWV was calculated (see Table II). It is observed that the two-and three-channel ratio models are better than NIR PWV. The RMSE values of NIR PWV are 3.83 mm, while the two-and three-channel ratio PWV values are less than 1.29 mm. Furthermore, independent ERA5

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TABLE II

ACCURACY INFORMATION OF TWO-AND THREE-CHANNEL RATIO PWV AND NIR PWV COMPARED WITH THE GNSS PWV IN ALL SAMPLES

Index	Type		
	2-Channel Ratio	3-Channel Ratio	NIR
RMSE (mm)	1.29	1.20	3.83
STD (mm)	1.29	1.20	2.08
Bias (mm)	0.02	0.02	3.17

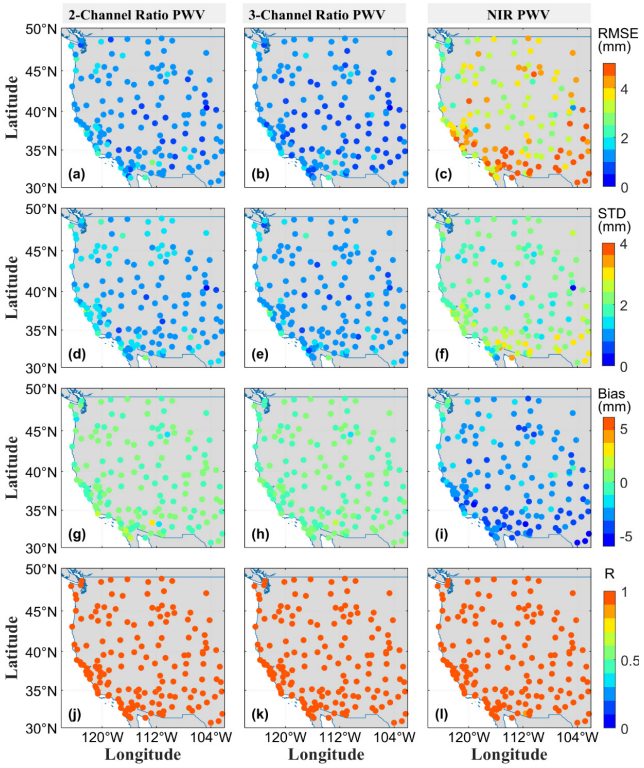


Fig. 9. (a)–(c) RMSE, (d)–(f) STD, (g)–(i) bias, and (j)–(l) *R* scores of the two-and three-channel ratio PWV and NIR PWV at individual sites compared with GNSS PWV.

PWV is used to verify the performance of the retrieval model (see Fig. S1). It is observed that the two-/three-channel ratio PWV is in good agreement with ERA5 PWV, and *R*/RMSE/STD/bias are 0.95/1.95 mm/1.9 mm/−0.46 mm

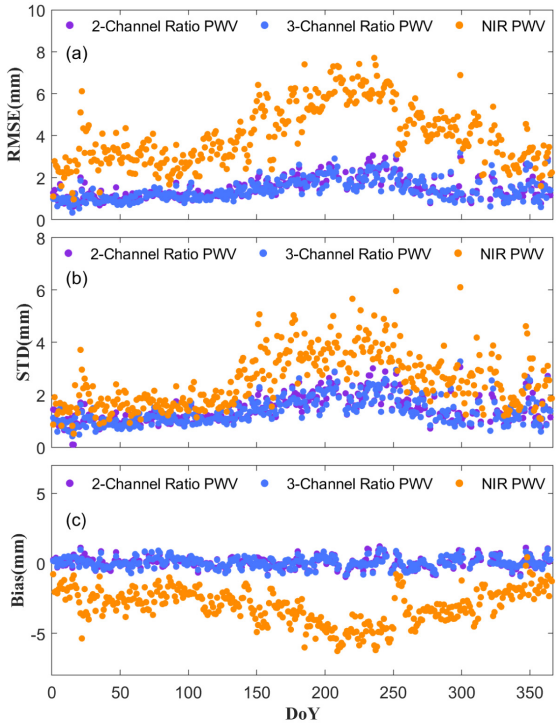


Fig. 10. (a) RMS, (b) STD, and (c) bias sequence for the two-and three-channel ratio PWV and NIR PWV compared with GNSS PWV.

and 0.96/1.9 mm/1.87 mm/−0.34 mm, respectively. NIR PWV is overestimated, *R*/RMSE/STD/bias of NIR PWV is 0.95/3.97 mm/2.77 mm/2.84 mm, and the two-/three-channel ratio model is better than NIR PWV. The above-mentioned results show that the rms of PWV retrieved by the proposed model is less than 3 mm, indicating that the two-/three-channel ratio PWV can be applied to the research of meteorological disasters [54].

B. Spatiotemporal Characteristics

The performance of the two-and three-channel ratio retrieval models based on BPNN at individual sites was verified. Fig. 9 shows the spatial distribution of RMSE, STD, bias, and *R* scores for the two-and three-channel ratio PWV and NIR PWV compared with those for GNSS PWV. Fig. 9(a)–(c) shows the RMSE generated by three PWV versus that

$$\text{Bias} = \frac{1}{N}\sum_{i=1}^{N}\left(\text{PWV}_i^{\text{pre}} - \text{PWV}_i^{\text{obs}}\right) \tag{12}$$

$$\text{STD} = \sqrt{\frac{1}{N}\sum\nolimits_{i=1}^{N}\left(\text{PWV}_i^{\text{pre}} - \text{PWV}_i^{\text{obs}} - \text{Bias}\right)^2} \tag{13}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum\nolimits_{i=1}^{N}\left(\text{PWV}_i^{\text{pre}} - \text{PWV}_i^{\text{obs}}\right)^2} \tag{14}$$

$$R = \frac{\sum_{i=1}^{N}\left(\text{PWV}_i^{\text{pre}} - \overline{\text{PWV}_i^{\text{pre}}}\right)\left(\text{PWV}_i^{\text{obs}} - \overline{\text{PWV}_i^{\text{obs}}}\right)}{\sqrt{\sum_{i=1}^{N}\left(\text{PWV}_i^{\text{pre}} - \overline{\text{PWV}_i^{\text{pre}}}\right)^2 \sum_{i=1}^{N}\left(\text{PWV}_i^{\text{obs}} - \overline{\text{PWV}_i^{\text{obs}}}\right)^2}} \tag{15}$$

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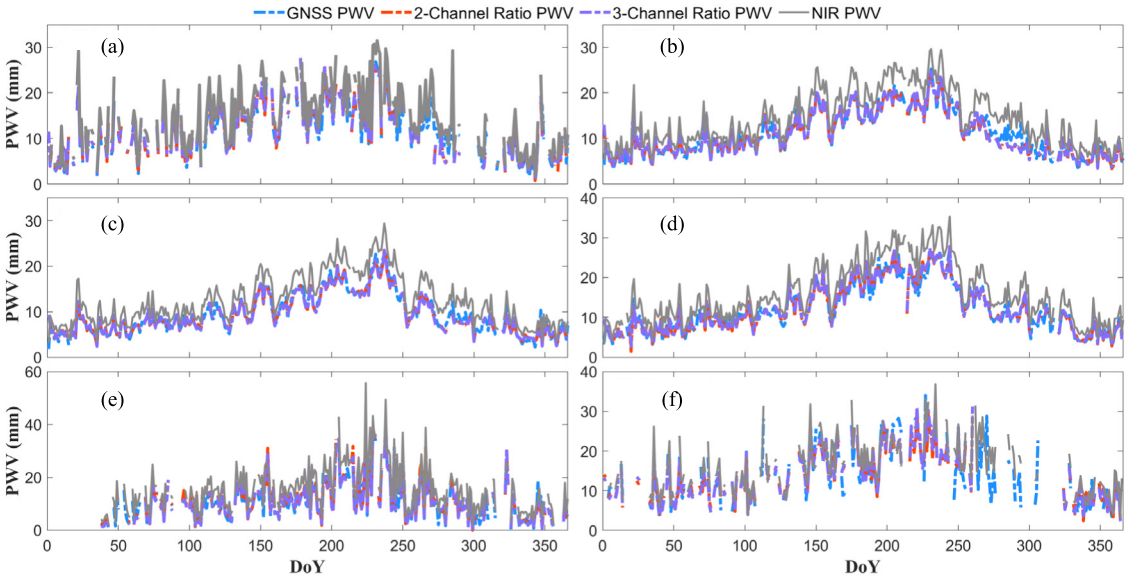


Fig. 11. Time series of GNSS PWV, two-channel ratio PWV, three-channel ratio PWV, and NIR PWV in different land-cover types during 2020. (a) Farmland. (b) Woodland. (c) Grassland. (d) Bare land. (e) Urban area. (f) Water body.

generated by GNSS PWV. It is observed that the RMSE of NIR PWV is higher than that of the two-and three-channel ratio PWV in North America, with a maximum RMSE being higher than 5 mm in the southern part of North America. While the RMSEs of the two-and three-channel ratio models are smaller than 2 mm in most areas, the RMSE improvements are 66.32% and 68.67%, respectively. Fig. 9(d)–(f) shows the STD of the three PWV values. The improvements in the STD generated by the two-and three-channel ratio models were significant, and the STD improvements were 37.98% and 42.31%, respectively. Fig. 9(g)–(i) shows the bias generated by GNSS PWV minus three PWV products at individual sites. The bias of the two-and three-channel ratio models at individual sites is equal to 0, while the NIR PWV shows a negative bias, which indicates that NIR PWV is overestimated. Fig. 9(j)–(l) shows the R scores between the two-/three-channel ratio/NIR PWV and GNSS PWV. R score exceeded 0.95 for the three PWV products. ERA5 PWV is also used to verify the proposed method (see Fig. S2). The result is similar to that of GNSS. It is observed that a good agreement between the two-and three-channel ratio PWV and ERA5 PWV can be found, while NIR PWV is poor. In the whole region, the improvements of rms and STD are 50.88%/52.14% and 31.41%/32.49%, respectively [see Fig. S2(a)–(f)]. Bias is also well-modified, while the bias of NIR PWV is obvious [see Fig. S2(g)–(i)], especially in the south of North America. R score is more than 0.95 for the three PWV products [see Fig. S2(j)–(l)]. Overall, the proposed methods are robust; they improve the accuracy of MODIS NIR PWV and are suitable for North America.

Fig. 10 shows the time sequences of rms, STD, and bias for the two-and three-channel ratio PWV and NIR PWV, where the purple circle is the two-channel ratio PWV, the blue circle is the three-channel ratio PWV, and the orange circle is the NIR PWV. Fig. 10(a) shows that the RMSE of NIR PWV has a notable temporal pattern, which initially increases and subsequently decreases, reaching a peak in summer. The RMSE time sequences of the two-and three-channel ratio PWV were weak. Moreover, the improvement of the two-and three-channel ratio PWV is especially obvious in summer. The STD sequences showed the same temporal pattern as the RMSE, reaching the highest value in summer [see Fig. 10(b)]. The STD improvements of the two-and three-channel ratio are weaker than that of RMSE, similar to the results in Fig. 9(d)–(f). Fig. 10(c) shows the bias sequences. The bias of the two-and three-channel ratio model has no apparent temporal model and fluctuates steadily around 0, indicating that the two-and three-channel ratio methods based on BPNN are the unbiased models with reference to GNSS PWV in this study. NIR PWV shows apparent negative bias and is overestimated at all times. Furthermore, validation with ERA5 also shows similar results, and details can be seen in Fig. S3. Overall, Figs. 9, 10, S2, and S3 show that the accuracy of the two-and three-channel ratio models can be guaranteed at any time and in any space.

Table III summarizes the verification of PWV obtained by the three methods for the six land-cover types. The results further show that PWV retrieved by the two-and three-channel ratio based on BPNN is robust for all land-cover types. The RMSE of the two-and three-channel ratio PWV was better than that of MODIS NIR PWV. Farmland had the best accuracy among all land-cover types. The RMSEs of the two-and three-channel ratio methods decreased to 1.05 and 0.99 mm, respectively. The accuracy of the two-and three-channel ratio methods is worst in the water body category; however, it is also within 2 mm, and the RMSEs are 1.73 and 1.61 mm, respectively. Water bodies sometime act as a black surface, affecting the signal of the absorption channel; therefore, the uncertainty of atmospheric scattering on the water body is larger [13], [25]. However, compared with NIR PWV, there are still improvements, with values of 44.37% and 48.23%,

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TABLE III

ACCURACY INFORMATION OF TWO-AND THREE-CHANNEL RATIOS AND NIR PWV IN DIFFERENT LAND-COVER TYPES. THE NUMBER IN PARENTHESIS UNDER LAND COVER INDICATES THE NUMBER OF SAMPLES

Land-cover types		Dataset	RMSE (mm)	STD (mm)	Bias (mm)	R
Farmland (386)	LC1	NIR	3.09	1.73	-2.56	0.98
		2-Channel Ratio	1.05	1.01	-0.02	0.99
		3-Channel Ratio	0.99	0.99	-0.06	0.99
Woodland (8475)	LC2	NIR	3.90	2.11	-3.26	0.97
		2-Channel Ratio	1.25	1.19	0.01	0.98
		3-Channel Ratio	1.10	1.03	0.01	0.98
Grassland (15207)	LC3	NIR	3.73	2.01	-3.97	0.96
		2-Channel Ratio	1.27	1.16	0	0.97
		3-Channel Ratio	1.18	1.08	-0.03	0.98
Bare Land (3362)	LC4	NIR	4.25	2.37	-3.52	0.97
		2-Channel Ratio	1.29	1.13	0.18	0.99
		3-Channel Ratio	1.18	1.09	0.12	0.99
Urban area (401)	LC5	NIR	5.05	2.79	-4.18	0.95
		2-Channel Ratio	1.58	1.56	-0.21	0.96
		3-Channel Ratio	1.56	1.56	0.03	0.97
Water body (442)	LC6	NIR	3.11	2.00	-1.72	0.96
		2-Channel Ratio	1.73	1.73	0.01	0.97
		3-Channel Ratio	1.61	1.60	0.11	0.97

respectively. In addition, the two-and three-channel ratio methods have the largest improvement in the urban area with RMSE values of 1.58 and 1.56 mm, respectively, and improvements of 68.71% and 69.31%, respectively. The STD of the two- and three-channel ratio methods has been improved for all land-cover types, but the improvement is weaker than RMSE. The bias equals zero because BPNN is an unbiased estimation model reference to GNSS PWV. In addition, we note that He and Liu [13] improved the MODIS NIR PWV retrieval algorithm for different surface types in North American land. We find that the accuracy of the proposed methods is much better than He and Liu [13] in six surface types, especially in urban and water areas. Overall, the two-and three-channel ratio retrieval methods based on BPNN showed satisfactory performance on different surfaces.

C. PWV Results Derived Using the Four Methods

In this study, the time sequences of BPNN-based two- and three-channel ratios, GNSS, and NIR PWV in six land-cover types are plotted in Fig. 11. It was observed that the values of the four PWV products were high in summer and low in winter. In addition, NIR PWV (gray lines) is in poor agreement with GNSS PWV, while the two-and three-channel ratio PWV have suitable consistency. It is worth noting that the consistency between the two-/three-channel ratio PWV and GNSS PWV in the water body is worse than those of the other five land-cover types, consistent with the results shown in Table III.

To compare the spatial distribution consistency of the four PWV products, Fig. 12 shows maps of GNSS PWV, MODIS NIR PWV, two-channel ratio PWV, and three-channel ratio PWV at 18:00 (UTC) on the 227th day of 2020. The spatial structure of the two-and three-channel ratio PWV is more similar to that of GNSS, while NIR PWV overestimates PWV. PWV in the red box indicates this point. In Fig. 12(b), MODIS PWV in the red box is larger than GNSS PWV, indicating that MODIS PWV may be inaccurate. The two-and three-channel ratio PWV in the red box is less than MODIS NIR

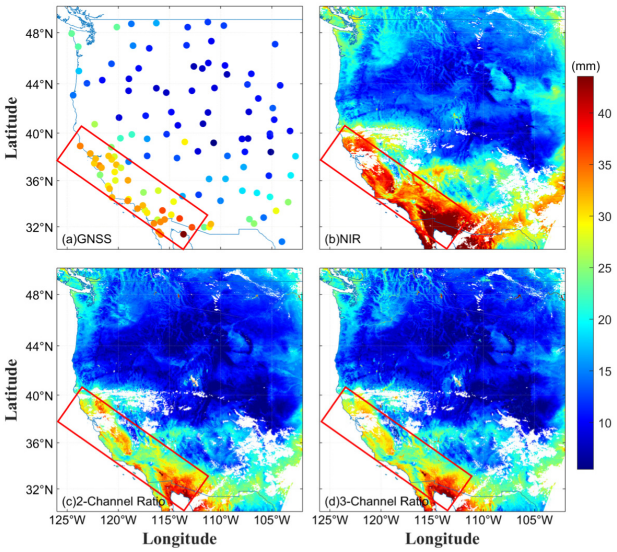


Fig. 12. Spatial distribution of (a) GNSS PWV, (b) NIR PWV, (c) two-channel ratio PWV, and (d) three-channel ratio PWV on 18:00(UTC) of the 227th day of 2020.

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PWV and close to GNSS PWV, indicating that the BPNN model is improved over MODIS NIR PWV. These results were consistent with the conclusions described earlier..

V. Conclusion

The MODIS sensor onboard satellite can observe the transmittance of the NIR channel and then retrieve the PWV. The traditional PWV retrieval algorithm uses a radiation transfer model to simulate the functional relationship between PWV and water vapor transmittance and constructs a model to obtain PWV information widely. Compared with high-precision ground-based PWV measurements, MODIS PWV was systematically overestimated. This study introduces high-precision ground-based GNSS PWV to construct a nonlinear function relationship between NIR atmospheric water vapor transmittance and GNSS PWV based on BPNN for different land-cover types, and a new retrieval algorithm considering land-cover types is proposed. As the two-and three-channel ratios can be used to calculate the atmospheric water vapor transmittance, two BPNN retrieval models were constructed. Taking GNSS PWV as the reference, the RMSE/STD/bias of MODIS NIR PWV products is 3.83/2.08/3.17 mm, while the RMSE/STD/bias values of PWV generated by the two-and three-channel ratios are 1.29/1.29/0.02 mm and 1.20/1.20/0.02 mm, respectively. The rms improvements of the proposed methods were 66.32% and 68.67%, and the average STD improvements were 37.98% and 42.31%, respectively. Moreover, ERA5 PWV is used to verify the performance of the proposed method, and the results are similar to that of GNSS. In addition, three PWV products are verified on different surfaces, and the results show that the two-and three-channel ratio methods can obtain satisfactory estimated PWV at any land-cover type. It should be emphasized that the BPNN-based three-channel ratio method is superior to the two-channel ratio results.

Compared with previous work, the method proposed in this study achieves satisfactory results. The relationship between atmospheric water vapor transmittance and PWV is complex and nonlinear, which is difficult to fit using traditional methods; however, the ANN technology brings new solutions to address problems. This study also shows that it is feasible and meaningful to use sparse high-quality ground-based observations combined with high-density remote sensing data to retrieve quantitative remote sensing parameters, which is key to improving the accuracy of quantitative remote sensing parameters.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this article.

Data Availability Statement

The data presented in this study are available upon request from the corresponding author.

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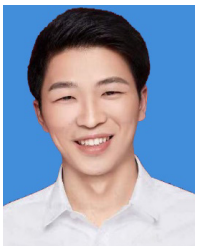
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